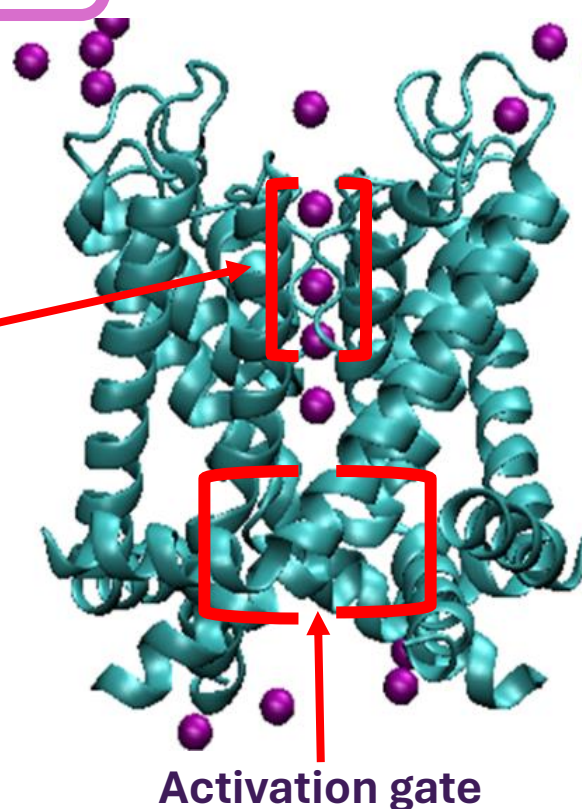


Tetraethylammonium Block of the Shaker Potassium Channel

Introduction

- The Shaker potassium channel is a **voltage-gated channel** whose main function is to reset the membrane potential after membrane depolarisation.
- The signature sequence of **selectivity filter** (SF) relates all potassium channels.
- Tetraethylammonium** (TEA): one of the most studied blockers but its binding mechanism remains unclear.
- Aims:** Understand TEA binding, its interactions with the channel, and its role in slowing down C-type inactivation.



Methods

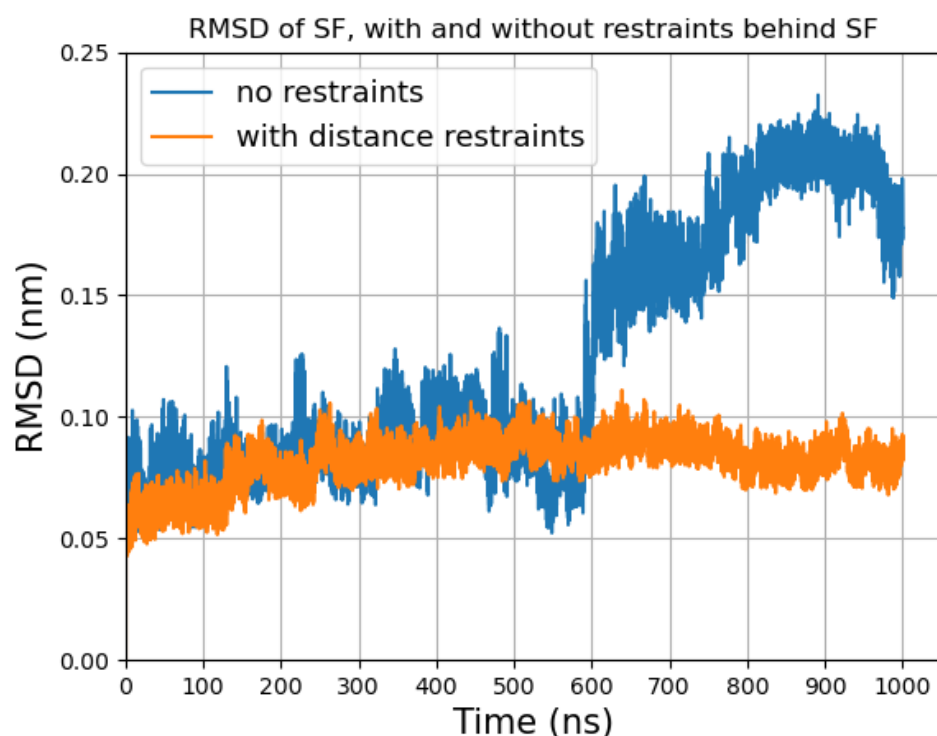
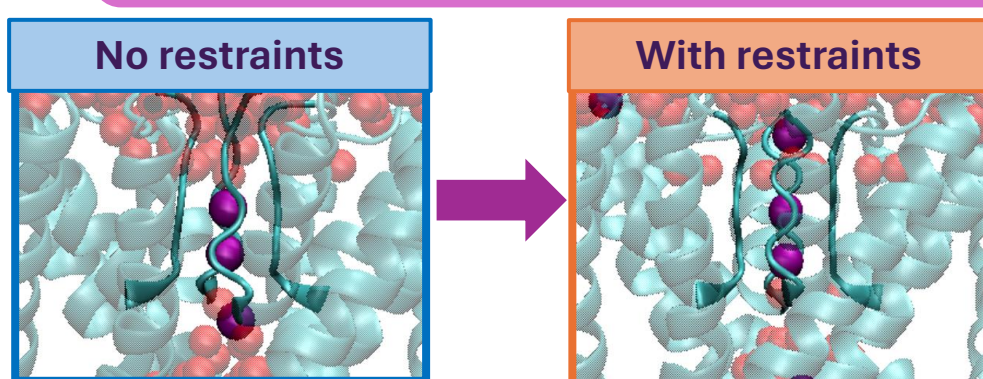
Shaker-IR potassium channel taken from RSCB Protein Data Bank – PDB ID: 7SIP.

Channel voltage-sensing domains removed to reduce system size for simulations.

CHARMM-GUI¹ embedded protein in POPC membrane and solvated with aqueous KCl.

Using **Molecular Dynamics** (MD) simulations with applied voltage

Broken Hydrogen Bond Distorts SF



Simulations without restraints show disruption of the hydrogen bond between **D447** and **W434** behind the SF.

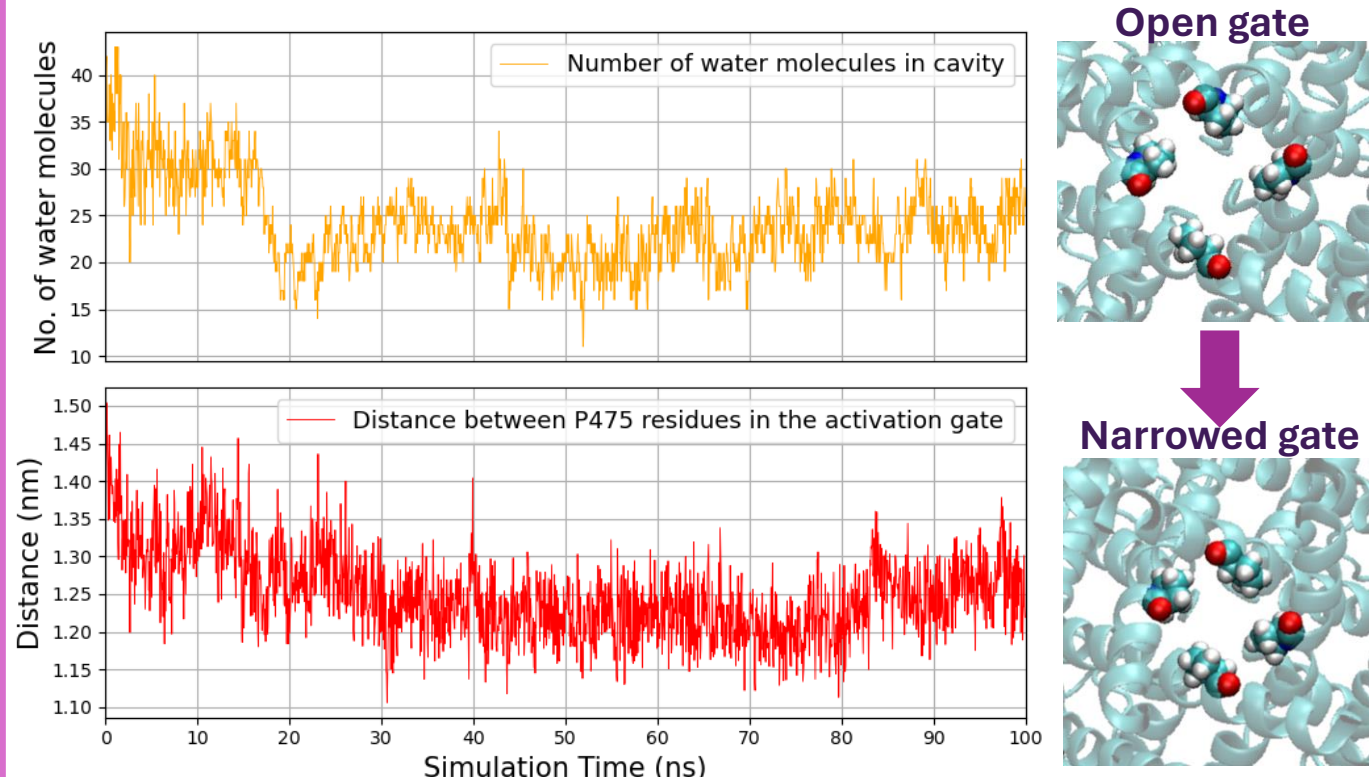
The SF is distorted, showing a significant change in structure for the channel with no restraints (**blue**).

These bonds are important in modulating slow (C-type) inactivation of the channel,² but this disruption is not expected to occur so quickly.

Distortions resolved by applying **distance restraints** between these residues. SF is now more stable (**orange**) to use in simulations with **TEA**.

Activation Gate Closing

First equilibration of the channel with TEA, internal TEA binding did not occur. We observed **activation gate narrowing** which precluded TEA could not access and bind within the cavity.

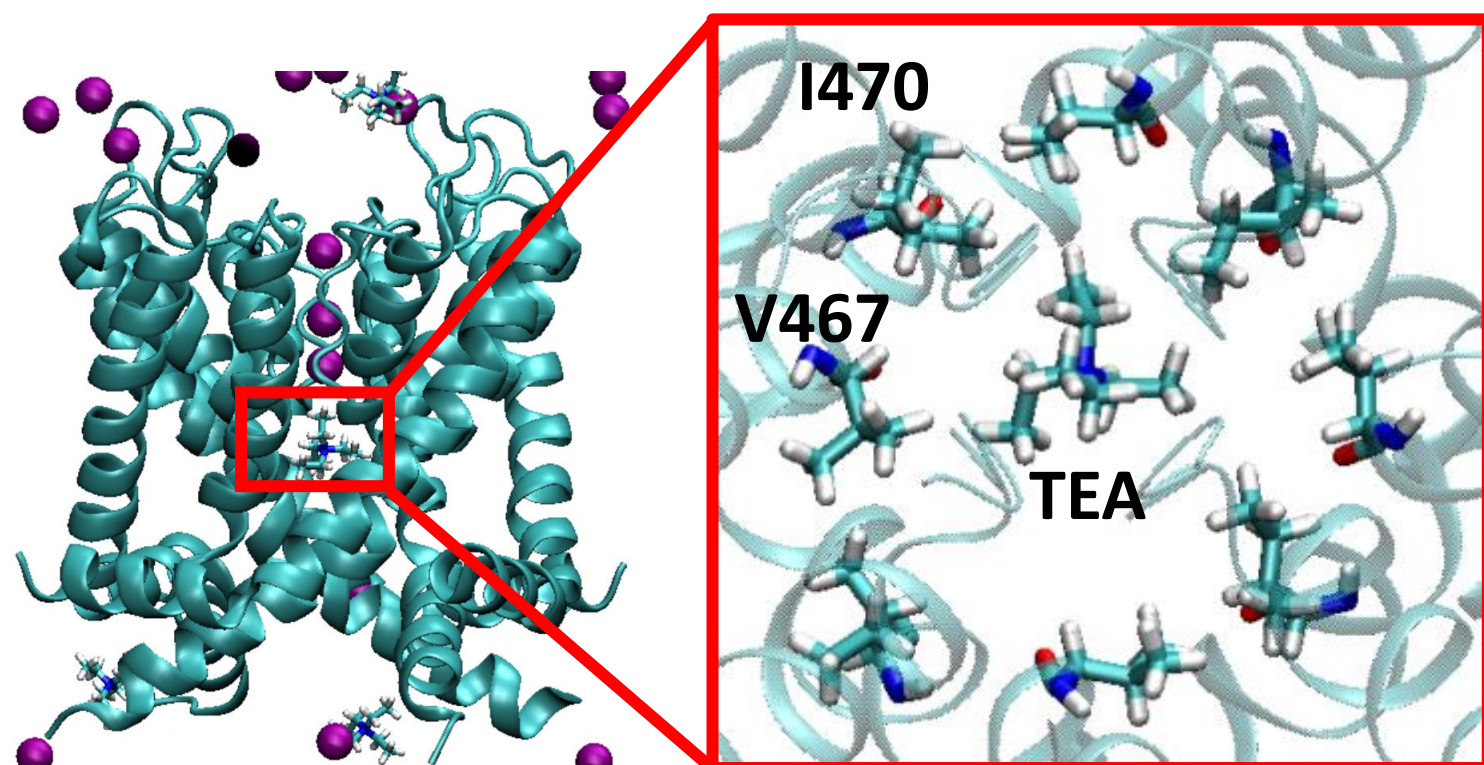


The distance between P475 residues in the gate was measured (**red**) to show the decrease. This gate narrowing is accompanied by **cavity dehydration** (**yellow**) (between SF and activation gate).

- Shown to be due to **non-polarizable CHARMM36m** forcefield³ not describing water-hydrophobic residue interactions properly

TEA Internal Binding Site

Equilibration with 300 mV applied voltage using a frame with the channel in an open conformation resulted in TEA binding **within** the cavity outside the SF.



The ethyl chains of TEA interact with a **hydrophobic ring** formed by residues of the inner transmembrane (TM) helices. (**I470** and **V467**)

- similar interactions were observed in a different K⁺ channel (KscA) between TEA and hydrophobic residue sidechains. These interactions also occurred with Ile residues⁴

Conclusion

- Hydrogen bonds between D447 and W434 are **required** for a stable SF.
- Internal binding of TEA involved **hydrophobic interactions** with the sidechains of residues in the inner transmembrane helices.

Future Work

- Production simulations with TEA in the system with +300 mV applied voltage are ongoing
- Simulations with no applied voltage and with -300 mV: to assess how **voltage** affects the **stability** of TEA below the SF
- Calculate the **free energy** of TEA binding
- Mutation** of residues at the activation gate (**P475D**). Mutating to aspartate should keep the gate in a stable open conformation

References

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